Introduction to Reinforcement Learning

Introduction

“Reinforcement Learning (RL) is a paradigm where intelligent agents learn to perform tasks by interacting with their environment. Unlike traditional AI systems that rely on hardcoded expert knowledge, RL agents acquire intelligence through experience—through successes and failures.”

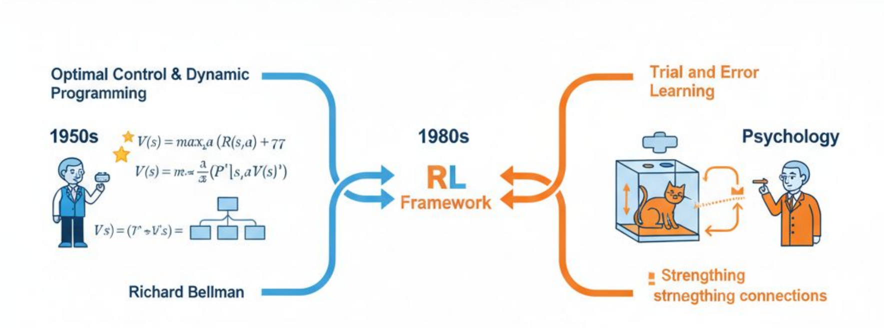
History:

- Optimal Control & Dynamic Programming (1950s)
- Learning by Trial and Error (Psychology)

Why It Matters Today:

The recent explosion of RL comes from combining DL.

- Play games at superhuman levels
- Control autonomous vehicles
- Optimize trading strategies
- Design new materials
- Assist in surgical procedures




missing salt?
Maybe more herbs?
ACTION: Add Spices/Salt
OBSERVATION/REWARD
PERFECT!

Intelligence and Artificial Intelligence

What is Intelligence?

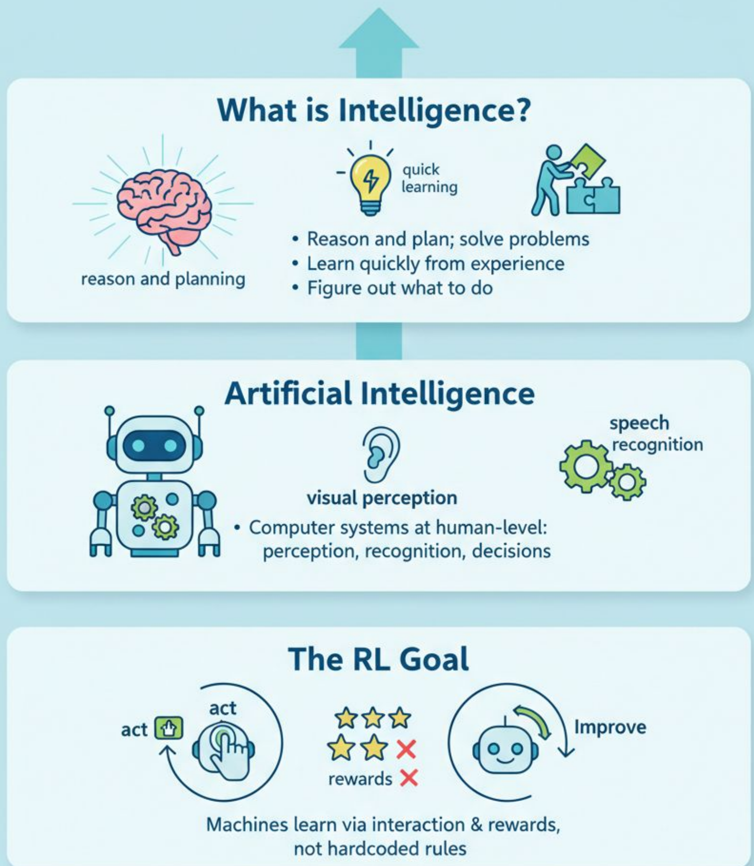
- Reason and plan; solve problems
- Learn quickly and learn from experience
- Ability to "figure out" what to do

Artificial Intelligence

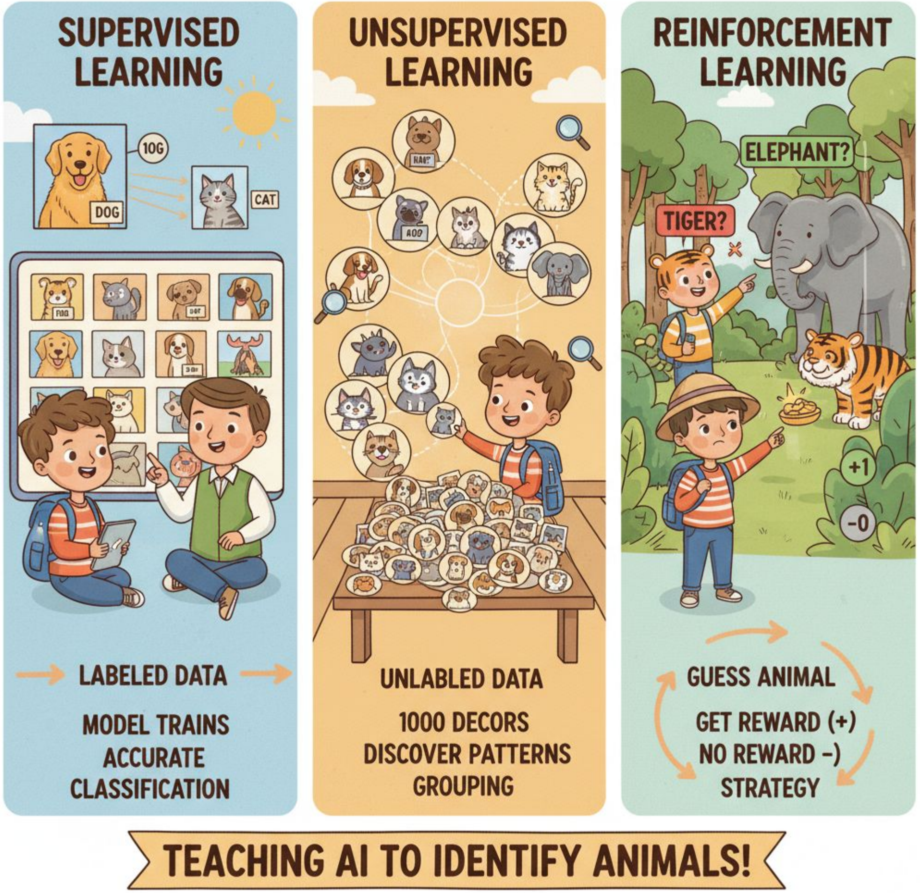
- Computer systems performing human-level tasks
- Visual perception, speech recognition, decision-making

The RL Goal

- Machines learning through interaction and rewards
- Not through hardcoded expert rules



Branches of Machine Learning



Branches of Machine Learning

Three Primary Branches:

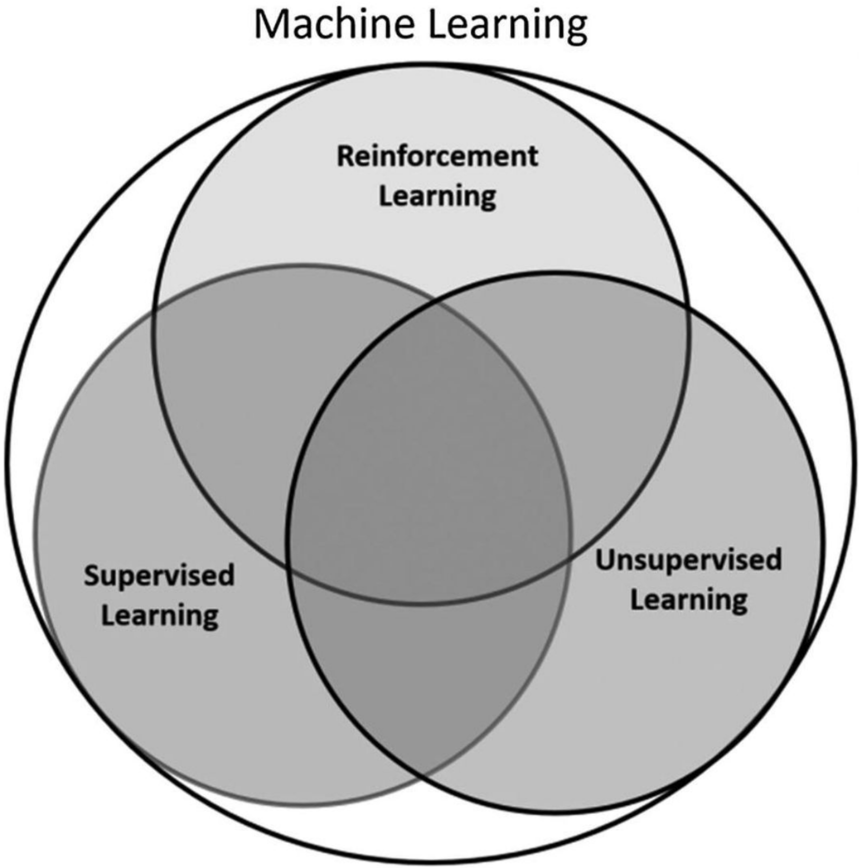
- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

Key Differentiator: Type and Timing of Feedback

- Supervised Learning: Labeled data - Predict labels for new data
- Unsupervised Learning: Unlabeled data - Discover hidden patterns
- Reinforcement Learning: Interaction Learn from delayed rewards

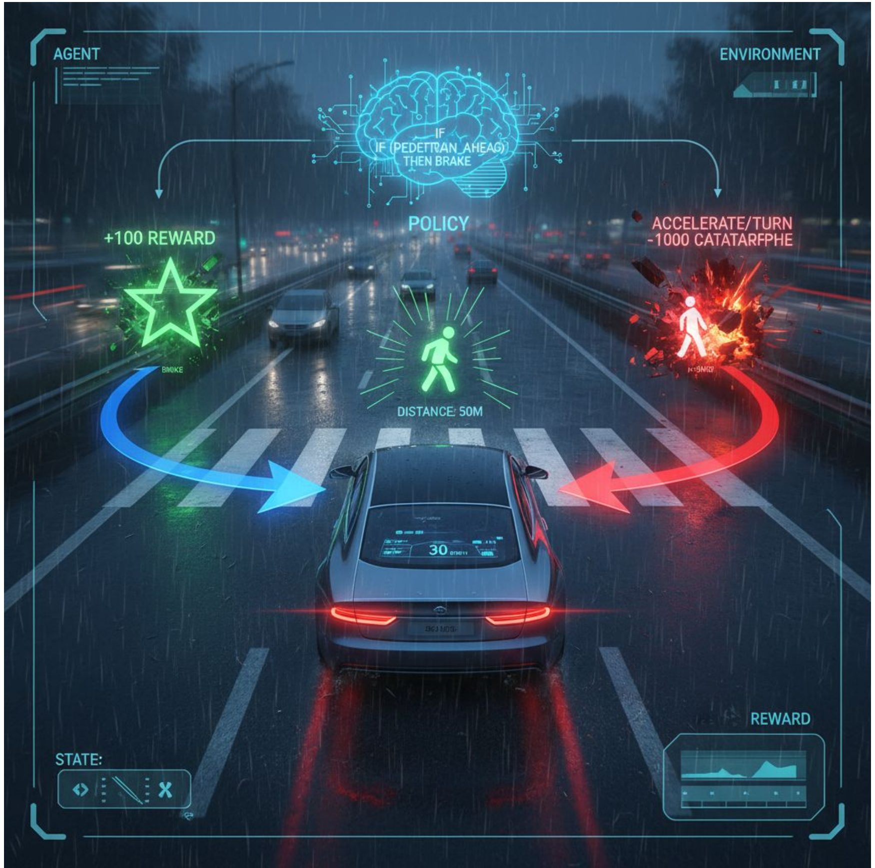
Why Three Approaches?

- Different problems require different learning signals
- Not all real-world problems have labeled training data
- Some problems require learning through experience



Why Not Just Supervised Learning?

Reinforcement Learning - Core Concepts



Reinforcement Learning - Core Concepts

Agent: The learner (car, robot, algorithm)

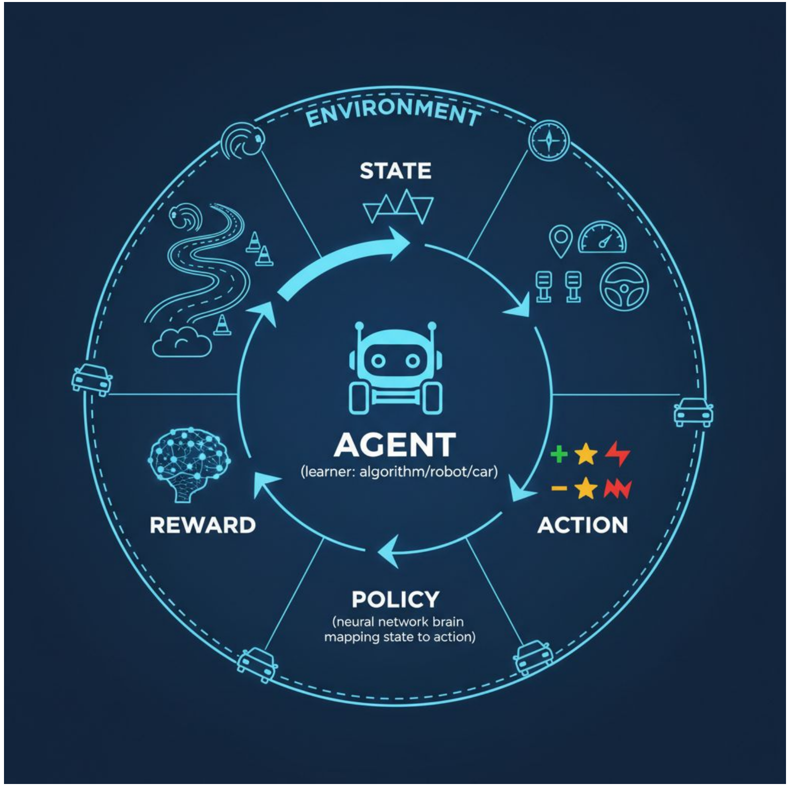
State: Current observation of the environment (position, speed, direction)

Action: Decision the agent makes (accelerate, brake, turn)

Reward: Feedback on action quality (positive for good, negative for bad)

Policy: Mapping from states to actions

Environment: Everything outside the agent's control



Formal Definitions

State (s): A complete description of the environment's current configuration.

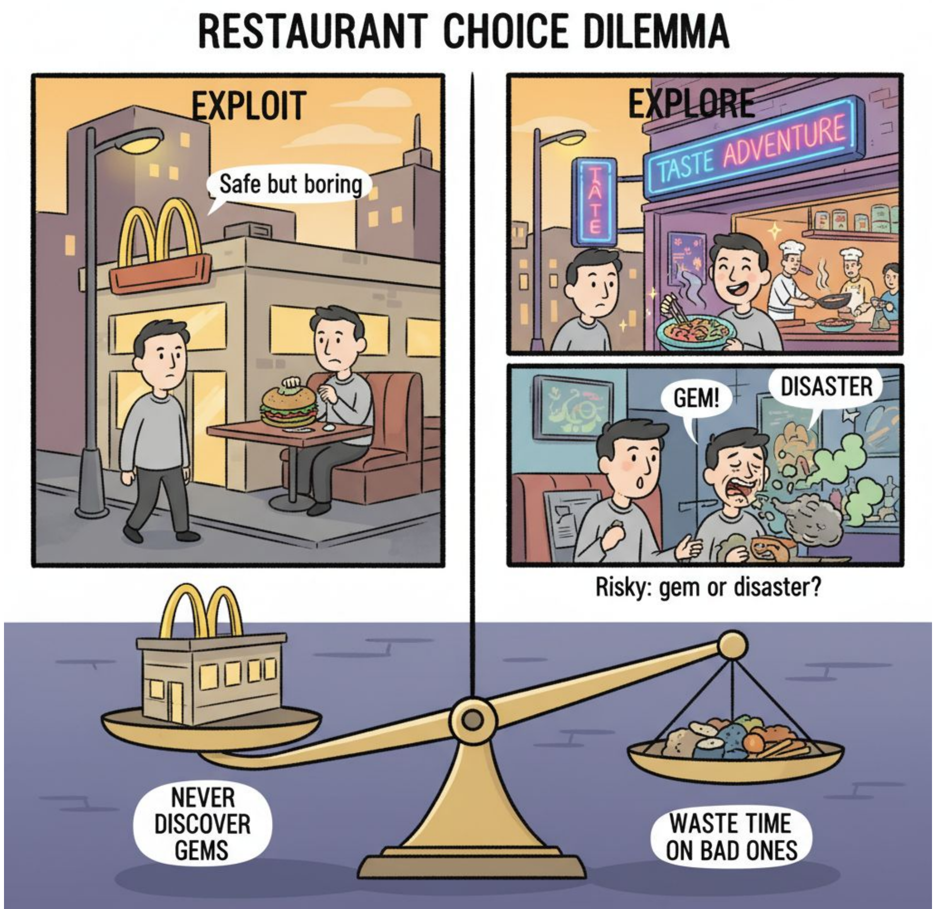
Action (a): A choice the agent makes while in a particular state. Actions have consequences—they determine the immediate reward and the next state.

Reward (r): A scalar value (number) the environment provides after the agent takes an action in a state. It's the agent's only signal about whether it's doing well or poorly. Key insight: Rewards are determined entirely by the environment (state + action), not by the agent's beliefs

Policy (π): The agent's decision-making strategy. It maps from states to actions. Formally, $\pi(a|s)$ = probability of taking action a in state s.

Value Function (V): The expected cumulative future reward starting from a particular state while following a specific policy.

The Exploration-Exploitation Dilemma



The Exploration-Exploitation Dilemma

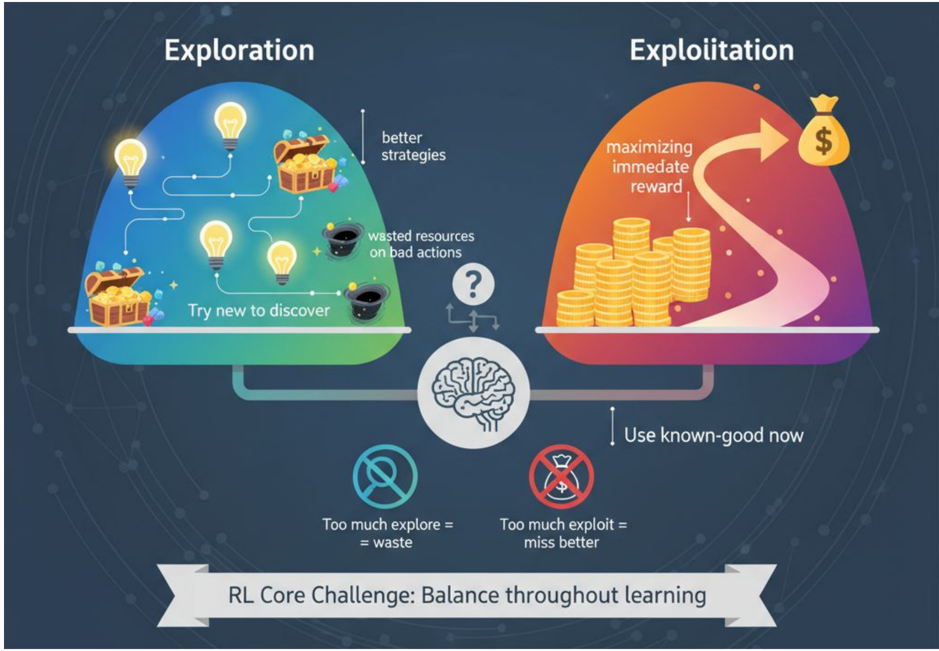
Exploration: Trying new actions to discover better strategies

Exploitation: Using known-good actions to maximize reward now

The Dilemma:

- Too much exploration = wasting resources on known-bad actions
- Too much exploitation = missing better strategies
- No universal rule for "enough exploration"

Core Challenge of RL: Balancing both throughout learning



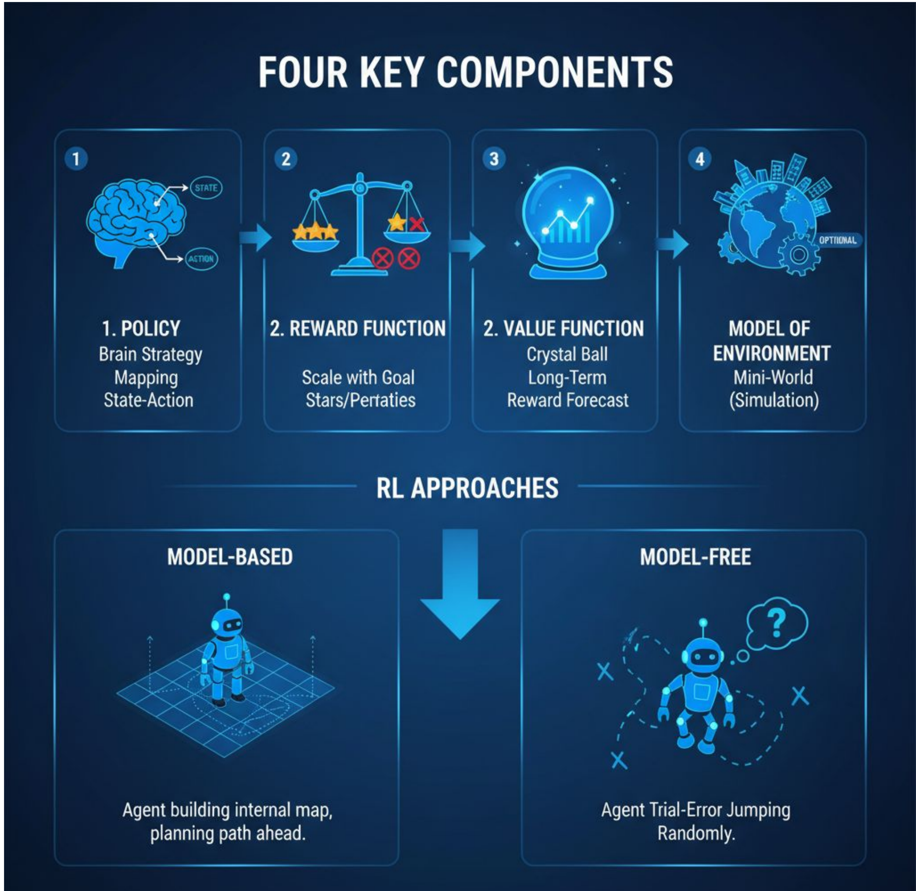
Core Components of RL Systems

Four Key Components:

1. Policy (the strategy)
2. Reward Function (the goal)
3. Value Function (long-term assessment)
4. Model of Environment (optional)

Two Approaches:

1. Model-based: Agent learns internal model, plans ahead
2. Model-free: Pure trial-and-error, no internal model



Deep Learning Meets Reinforcement Learning

Before 2014: RL limited to small state spaces

2014 Breakthrough: DeepMind combines Deep Learning + RL

Key Innovation: Deep Q-Networks (DQN)

Challenges Solved:

- Learning from raw sensory data (images, audio)
- Handling high-dimensional state spaces
- Scaling to complex real-world problems

Modern RL: Nearly all advanced algorithms use deep learning

Real-World Applications

Autonomous Vehicles

- Path planning and trajectory optimization
- Lane changing, overtaking, parking
- Uses inverse RL to learn from expert drivers

Robotics

- Learning gaits (how to walk)
- Object manipulation
- Imitating human behavior

Recommendation Systems

- Learning from user feedback
- Dynamic adaptation to changing preferences
- Examples: YouTube, TikTok, Facebook, E-commerce

Real-World Applications

Finance and Trading

- Time-series optimization
- Automated trading strategies
- Portfolio risk management
- Chatbots for user support

Healthcare

- Predictive medicine and early intervention
- Robot-assisted surgery
- Treatment optimization
- Dynamic imaging interpretation

Game Playing

- Board games (Go, Chess)
- Video games
- Complex simulations

Key Takeaways

What is RL: Agents learning optimal behavior through interaction and rewards

Why RL: Solves problems too complex for rule-based or supervised approaches

When to use RL: Sequential decision-making with delayed rewards

Core Challenge: Balancing exploration and exploitation

Modern RL: Deep learning enables learning from raw, complex data

Next Steps: Formal MDP framework, value functions, and algorithms

Quiz 1

In a basic reinforcement learning setup, what best describes the relationship between the agent and the environment?

A. The environment chooses actions, and the agent passively observes outcomes
B. The agent predefines a fixed sequence of states, and the environment assigns rewards
C. The agent chooses actions, the environment responds with next state and reward
D. Both agent and environment choose actions simultaneously at each time step

Quiz 2

What is the primary role of a policy in reinforcement learning?

A. To store all past rewards the agent has ever received
B. To map states (or observations) to actions the agent should take
C. To guarantee that the agent always explores new actions
D. To compute the exact dynamics of the environment's transitions

Quiz 3

Which scenario best illustrates the exploration vs exploitation trade-off in reinforcement learning?

A. An agent ignoring rewards and focusing only on matching a fixed action sequence
B. An agent only choosing completely random actions forever
C. An agent always repeating the highest-reward action it has found so far
D. An agent sometimes trying new actions even if they currently seem worse, to discover potentially better long-term rewards

Quiz 4

What is the main distinction between model-based and model-free reinforcement learning methods?

A. Model-based methods assume rewards are always zero, while model-free methods learn rewards
B. Model-based methods learn or use a model of environment dynamics, while model-free methods learn values or policies directly from experience
C. Model-based methods require deep neural networks, while model-free methods do not
D. Model-based methods search over all possible policies, while model-free methods cannot improve policies

Quiz 5

In a deep Q-network (DQN), what is the primary purpose of using a deep neural network?

A. To approximate the action-value function when the state space is too large or continuous for a table
B. To exactly compute the true optimal Q-values for all states and actions analytically
C. To remove the need for rewards by predicting states directly
D. To ensure the environment’s dynamics become deterministic